Complete CSS Study Guide

From beginner basics to advanced techniques — your comprehensive journey to mastering Cascading Style Sheets for modern web development.

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What is CSS?

CSS (Cascading Style Sheets) is the styling language that transforms plain HTML into visually appealing websites. It controls layout, colours, fonts, spacing, and animations.

Three Ways to Include CSS:

- **Inline:** style="color: blue;"
- **Internal:** <style> in HTML head
- **External:** separate .css file (recommended)

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CSS Selectors & Combinators

Basic Selectors

- **Element:** h1 { }
- **Class:** .nav { }
- **ID:** #header { }
- **Universal:** * { }

Combinators

- **Descendant:** div p
- **Child:** div > p
- **Sibling:** h1 + p
- **General:** h1 ~ p

Pseudo-Classes

- :hover, :focus
- :nth-child(2n)
- :first-child
- ::before, ::after

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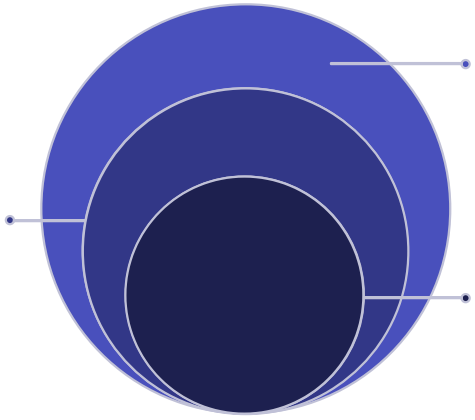
Understanding the Box Model

Every Element is a Box

The CSS box model defines how space around elements is calculated. From inside out: content, padding, border, margin.

Pro Tip: Use box-sizing: border-box to include padding and border in width calculations.



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Colours, Units & Typography

Colour Formats

- **HEX:** #FF5733
- **RGB:** rgb(255, 87, 51)
- **HSL:** hsl(9, 100%, 60%)

CSS Units

- **Absolute:** px, pt
- **Relative:** %, em, rem
- **Viewport:** vh, vw

Typography

- font-family, font-size
- font-weight, line-height
- text-align, letter-spacing

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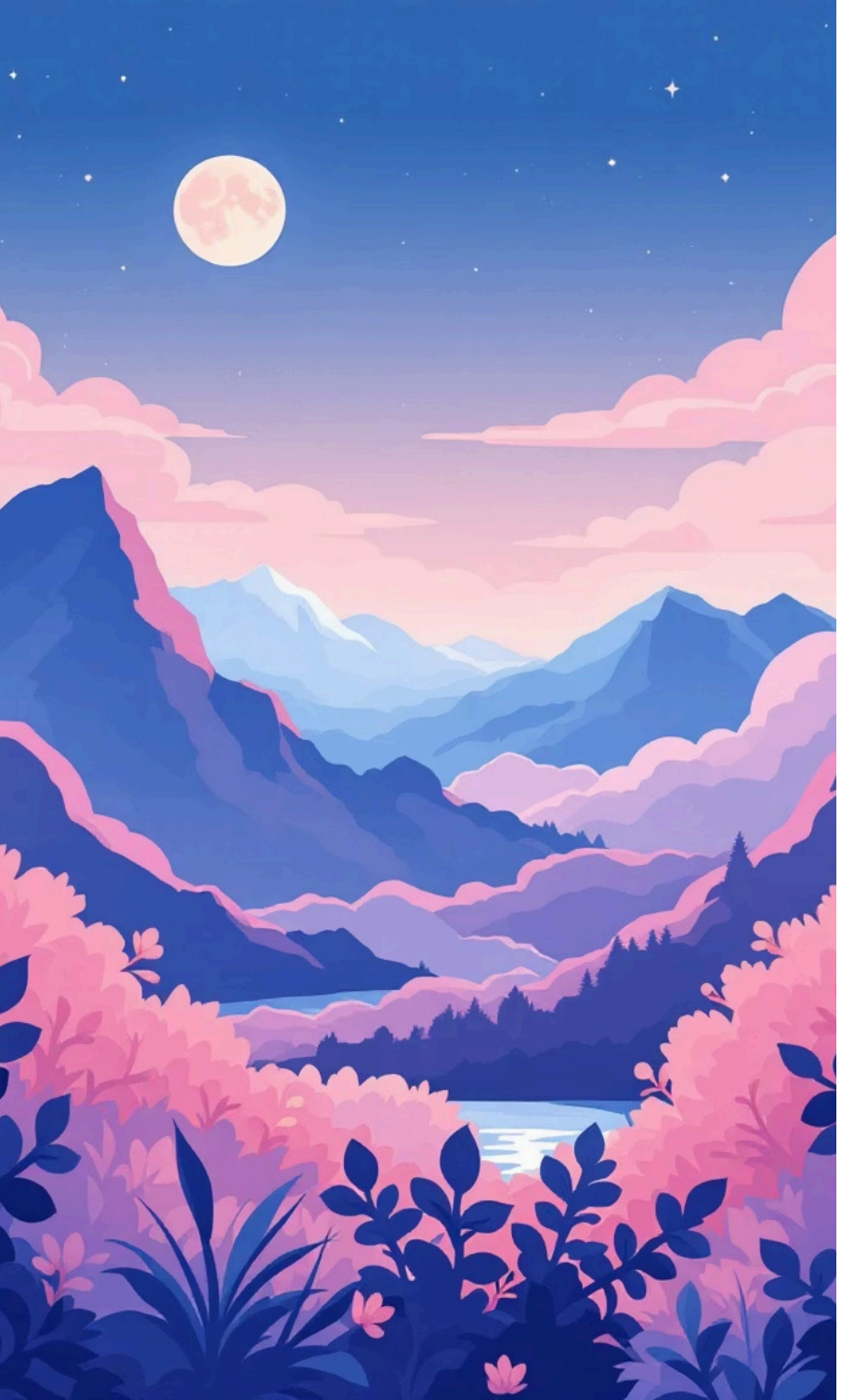

Positioning & Display Properties

Position Values

- **static:** default flow
- **relative:** offset from original
- **absolute:** positioned ancestor
- **fixed:** viewport relative
- **sticky:** scroll-based

Display Options

- **block:** full width
- **inline:** content width
- **inline-block:** best of both
- **flex:** flexible layouts
- **grid:** 2D layouts

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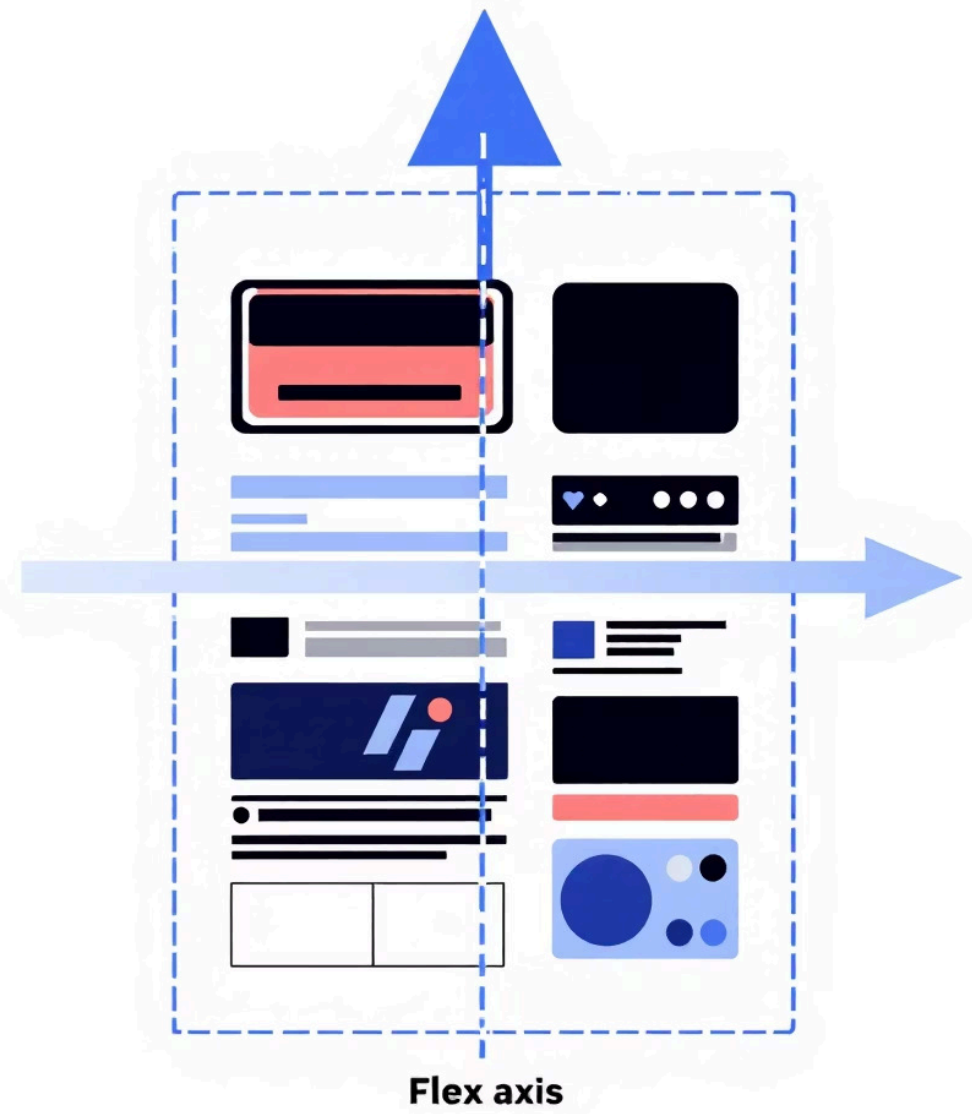
Flexbox: 1D Layout Mastery

Main Concepts

Flexbox arranges items along a single axis (main or cross). Perfect for navigation bars, button groups, and centering content.

- **justify-content:** main axis alignment
- **align-items:** cross axis alignment
- **flex-wrap:** wrapping behaviour
- **flex-grow:** space distribution



Flexbox is ideal for one-dimensional layouts and component-level design patterns.

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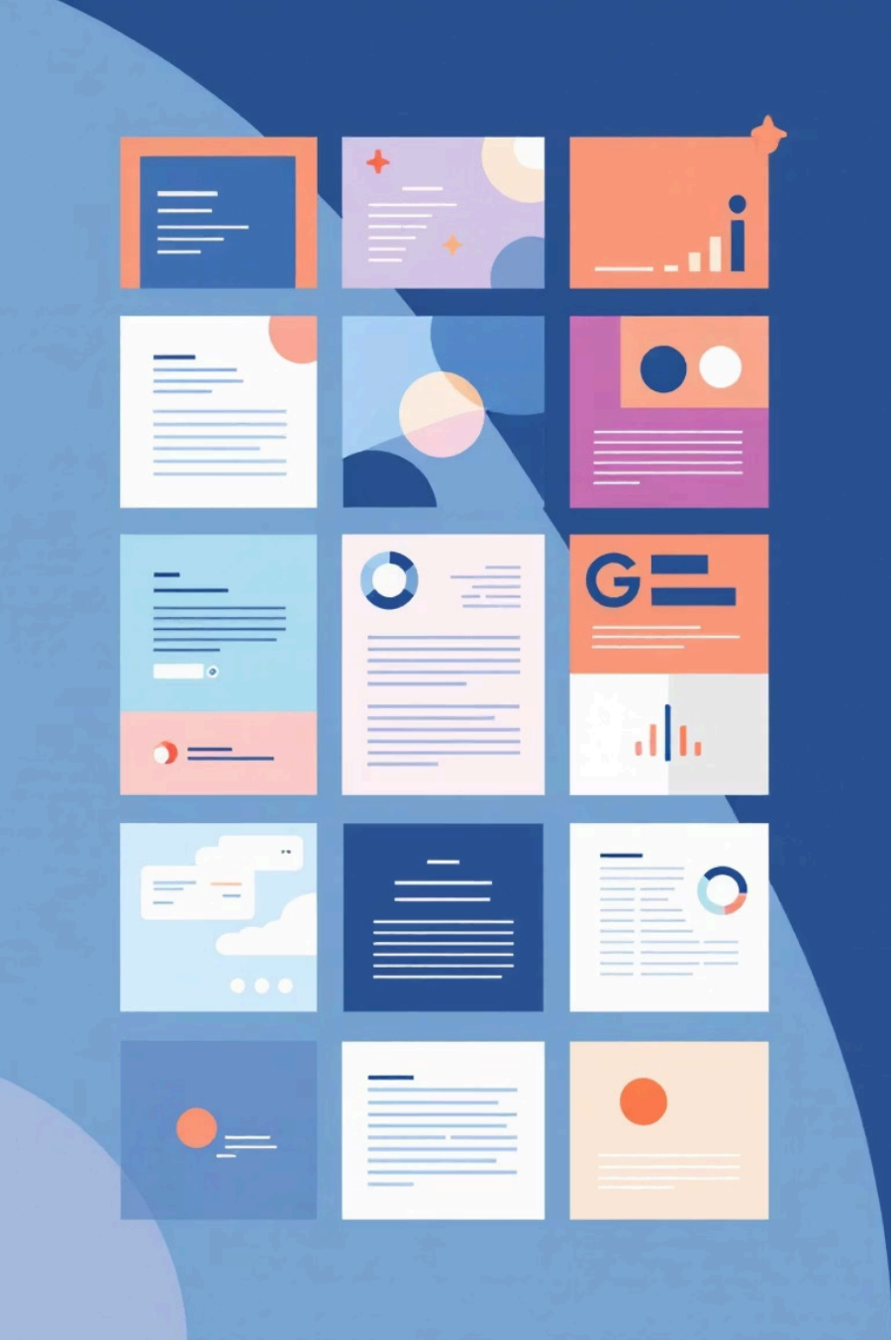

CSS Grid: 2D Layout Power

01

Define Grid Container

Set display: grid and define rows/columns using grid-template-rows and grid-template-columns.

02

Use the FR Unit

The fr (fraction) unit distributes available space. 1fr 2fr creates a 1:2 ratio between columns.

03

Position Grid Items

Use grid-area, grid-column, or grid-row to place items precisely within the grid structure.

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Responsive Design & Media Queries

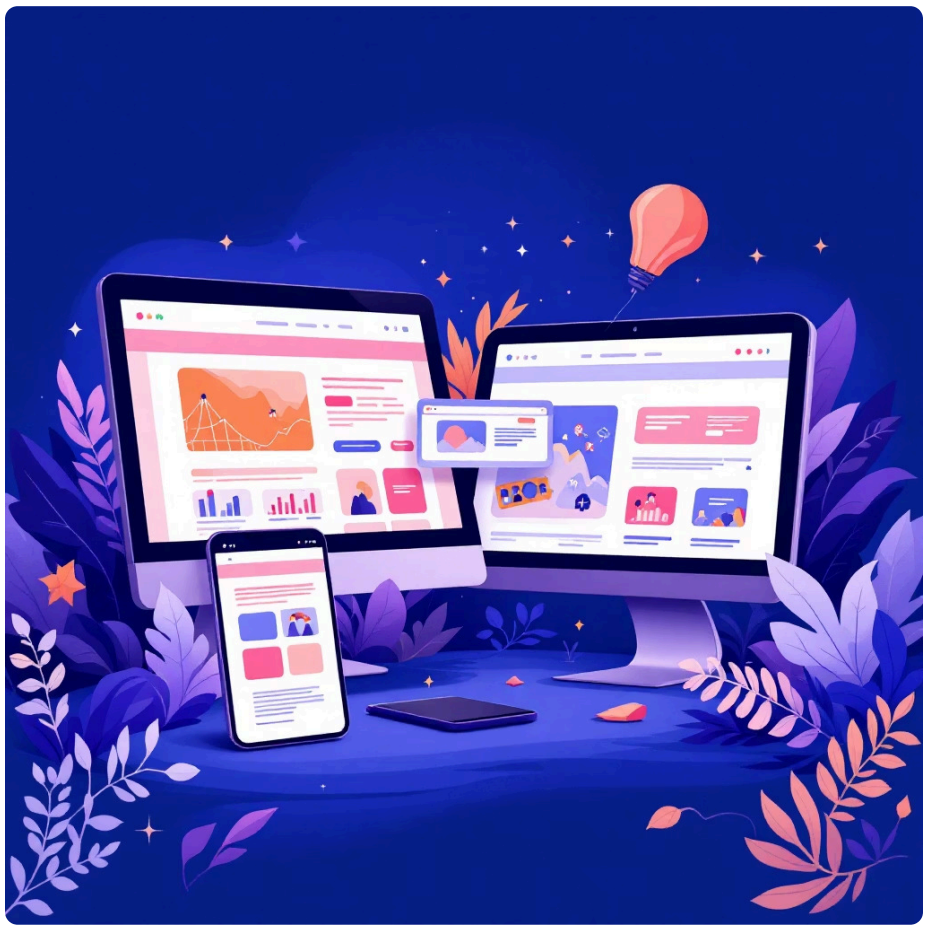
Mobile-First Approach

Start with mobile styles, then enhance for larger screens using media queries. This ensures optimal performance on all devices.

```
/* Mobile styles first */
.container { width: 100%; }

/* Tablet and up */
@media (min-width: 768px) {
  .container { max-width: 720px; }
}

/* Desktop and up */
@media (min-width: 1024px) {
  .container { max-width: 960px; }
}
```



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Advanced CSS & Best Practices

Animations

Use transitions for hover effects and @keyframes for complex animations. Control timing with easing functions.

CSS Variables

Define custom properties with --variable-name in :root, then use var(--variable-name) throughout your stylesheet.

Best Practices

Keep code DRY, use external stylesheets, follow BEM methodology, and leverage browser dev tools for debugging.

Master these fundamentals, then explore CSS frameworks like Bootstrap or Tailwind for rapid development. Continue learning with MDN, CSS-Tricks, and practical projects.

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